

# NOLAN A. ROBERTSON

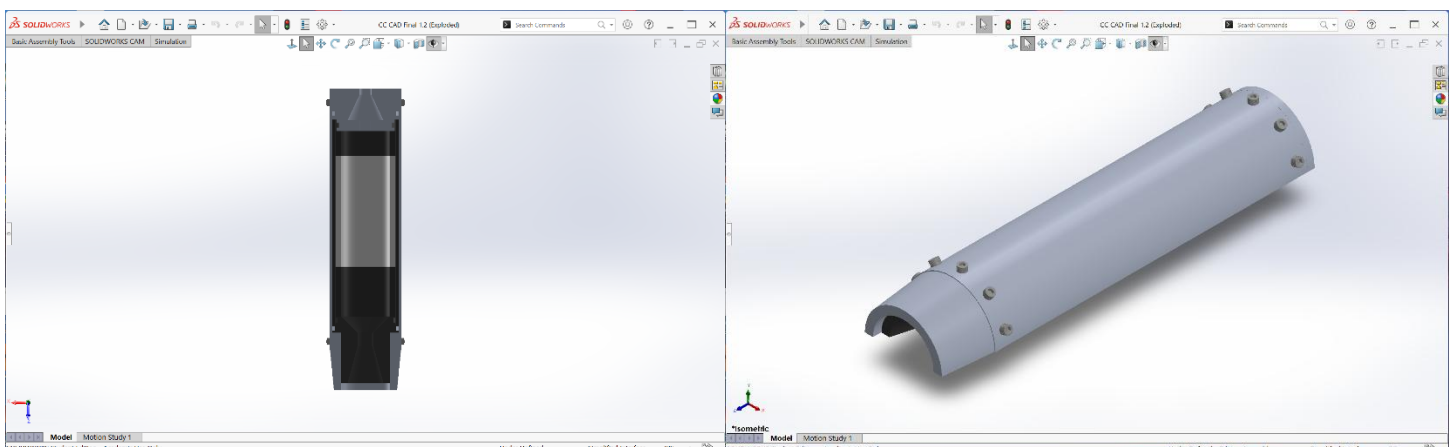
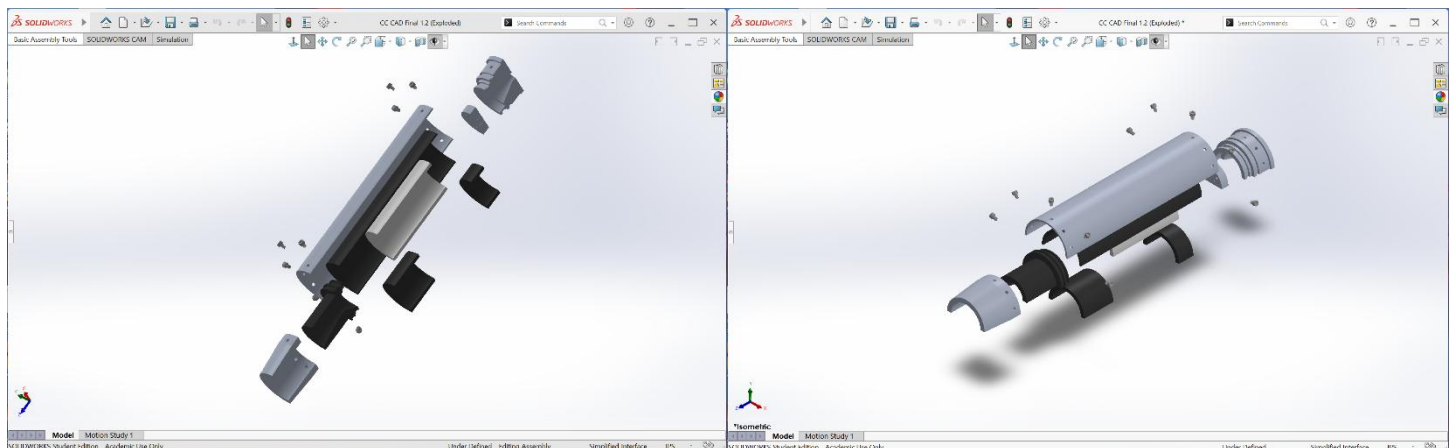
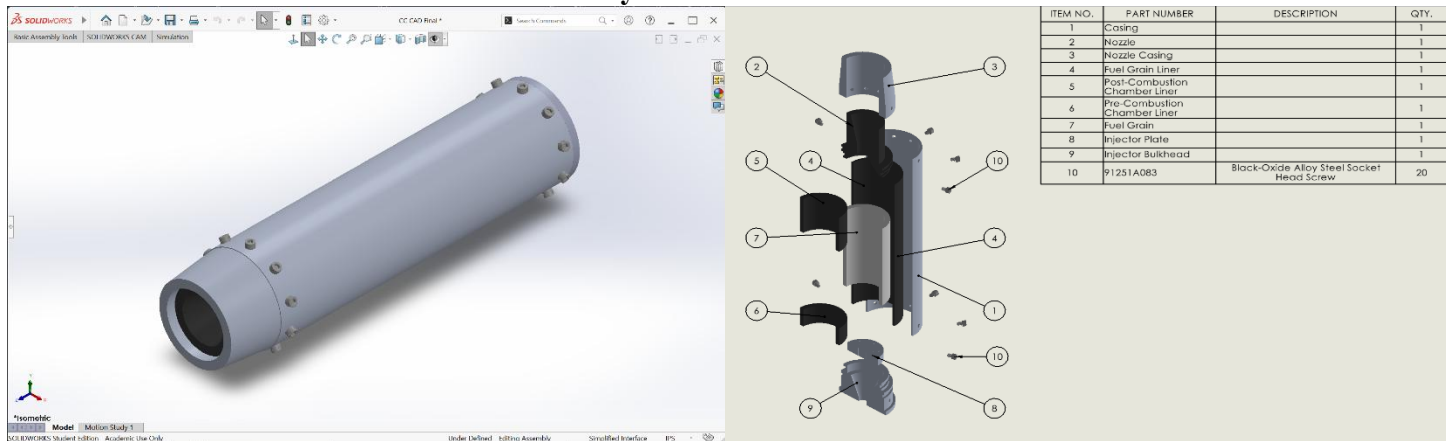
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## Combustion Chamber, Hybrid Rocket Motor N-class and K-class

Fall 2024-Fall 2025

- Contributed to the design and development of the combustion chamber of an N-class hybrid rocket motor (20,000 N-s of total impulse) Fall 2024 and Spring 2024
- Spring 2025 spearheaded the design of a K-class motor (2,500 N-s of total impulse). Made extensive and iterative design decisions regarding safety, thermal and structural analysis, optimization, material selection, fabrication, all within tight time and cost constraints, and in close coordination with several other design teams. Produced a complete CAD assembly for the K-class motor.

### K-class Combustion Chamber CAD Assembly:



- ## Project Poster:



Evan Daniel, Abel Saberhagen, Omer Mirza, Nolan Robertson, Jon Buszard, Austin Phan

## FUNCTION

## FINDINGS AND FUTURE

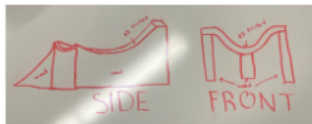
Our project was ultimately successful in addressing the issue of inaccessible bike parking by developing a functional bike parking notification system. If we had more time or better materials, we would use a waterjet to produce more precise parts, incorporate additional 3D printing for better cohesion within the system, and fully integrate the system with the website wirelessly to enhance real-time functionality.

1. Our design must be user-friendly and be effective for those traveling to classes.
2. Bicycle users must be able to easily see the availability of a parking space before they get to class.
3. The design must be compact enough to fit within standard bike parking areas
4. The sensor and notification system must work consistently
5. The system should significantly improve bike parking organization and increase convenience.

Wu, Y. (n.d.). Undergraduate Commuter Students' Challenges and Struggles. ERIC. Retrieved November 6, 2024, from <https://eric.spc.edu/wp-content/uploads/2024/05/1247618.pdf>

AASHE. (n.d.). University of Virginia OP-18: Commuter Model Split: The Sustainability Tracking, Assessment & Rating System. Retrieved November 6, 2024, from <https://reports.aashe.org/assessment/university-of-virginia-report/2024-03-04/C19commuterSplit/C19-18>

Bike Theft Prevention - Public Safety. (n.d.). Macalester College. Retrieved November 6, 2024, from <https://www.macalester.edu/bike-theft-prevention/bike-theft-prevention/>



## The image consists of two side-by-side photographs of a curved wooden structure, possibly a model of a ship's hull or a curved beam. The structure is made of light-colored wood and has a blue interior lining. It is shown from two different angles, highlighting its curved shape and the blue interior lining. The structure is placed on a wooden surface.

## Bridge

Fall 2025

- Designed and manufactured a “bridge” able to hold 50 lbf for a class competition utilizing 2 in<sup>3</sup> of 3D ABS plastic with a team of 6. Project involved the use of progressive design, CAD, material and structural analysis, and team coordination.

Our final design and SolidWorks FEA analysis

